

# Tomato stem SEM preparation

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1. Tissue cut to appropriate size (1-3mm<sup>2</sup>), (height 1-5mm) **\*All steps @RT**
2. Fixation
  - 1) 2.5% glutaraldehyde (0.1M phosphate buffer, pH 7.2-7.4) Vacuum 30min + O/N (hood)
  - 2) or 4% FA (with 1x PBS)
3. Washing the tissue with PBS buffer 20 min X 3 (hood)
4. extract cytoplasmic materials
  - 1% (v/v) Triton X-100 for **72 h** at 30°C or RT Shaker
  - Rinse the tissue with D.W x3
5. Post fixation
  - 1% Osmium tetroxide (0.1M phosphate buffer or 1M PBS) hood → O/N  
@darken space, 4°C
6. Washing the tissue with PBS buffer 20 min X 3 (hood)
7. Dehydration in increasing concentration of alcohol (EtOH)
  - 30, 50, 60, 70, 80, 90, 95, 100, 100, 100% at least 2hr  
100% step **O/N**